MIS 201 · WEEK 6 · MONDAY 10/26

Ch. 6: Bias: Hiring & Lending: The Machines That Judge Us

Part 1. A classifier learns who got the job or the loan, not who deserved it. How does old bias get into a new machine, and how would anyone check?

Part 2. Amazon's recruiting engine: built to find the best candidates, it taught itself to downgrade women, and Amazon scrapped it.

Then: 30 minutes with your Prototype Project team.

Due Friday 10/30: your project team & idea

FINAL PROJECT · 10%

WHAT

Four names, plus one paragraph on your idea: the problem, who has it, and what your AI system would do.

WHERE

On Canvas, one submission per team.

WHEN

Friday 10/30, 11:59 p.m.

NO TEAM YET?

The last 30 minutes of today are for teams. Anyone without a team by Friday gets placed on one.

Also due Friday 10/30, 11:59 p.m.: the Lab 6 writeup and the Chapter 6 quiz, both on Canvas.

Today

PART 1 · CHAPTER 6 CONCEPTS · ~40 MIN

1:50 a.m., and the classifier	~7 min
Bias, Gender Shades, and the objectivity trap	~10 min
Proxies, redlining, and the four-fifths rule	~9 min
Two kinds of wrong: the invisible pile	~5 min
How to check a screener: the audit	~5 min
Try it: what's the problem?	~3 min

PART 2 · THE AMAZON CASE · ~38 MIN

Background and what happened	~10 min
Amazon as an AI system	~5 min
Discuss in groups, then as a class	~18 min
Keep, fix, or drop	~5 min

PART 3 · PROTOTYPE PROJECT TEAMS · ~30 MIN

Find your team, pick a problem, draft the paragraph	~30 min

BY THE END, YOU CAN

- Say what a classifier learns, and what it doesn't
- Explain the objectivity trap
- Find a proxy on a real résumé
- Tell a false positive from a false negative, and why "95% accurate" can hide a lot
- Run a matched pair and the four-fifths rule

PART 1 · CHAPTER 6 · ABOUT 40 MINUTES

AI Bias: Hiring & Lending: who does the machine say no to?

Three questions for the week.

1. What does a classifier learn, and what does it never learn?
2. How does bias get in when nobody put it there?
3. When the no is wrong, who finds out, and who answers?

The point of the chapter: a classifier learns who got the job, not who deserved it, then applies that pattern to everyone, every time, at any hour of the night.

Derek Mobley: rejected at 1:50 a.m.

12:55 a.m.

Derek Mobley applies for a job online.

1:50 a.m.

Rejected, less than an hour later.

Who said no?

No person at the company read it. Résumé-sorting software did.

THEN, OVER SEVERAL YEARS

100+ jobs. Every one, no.

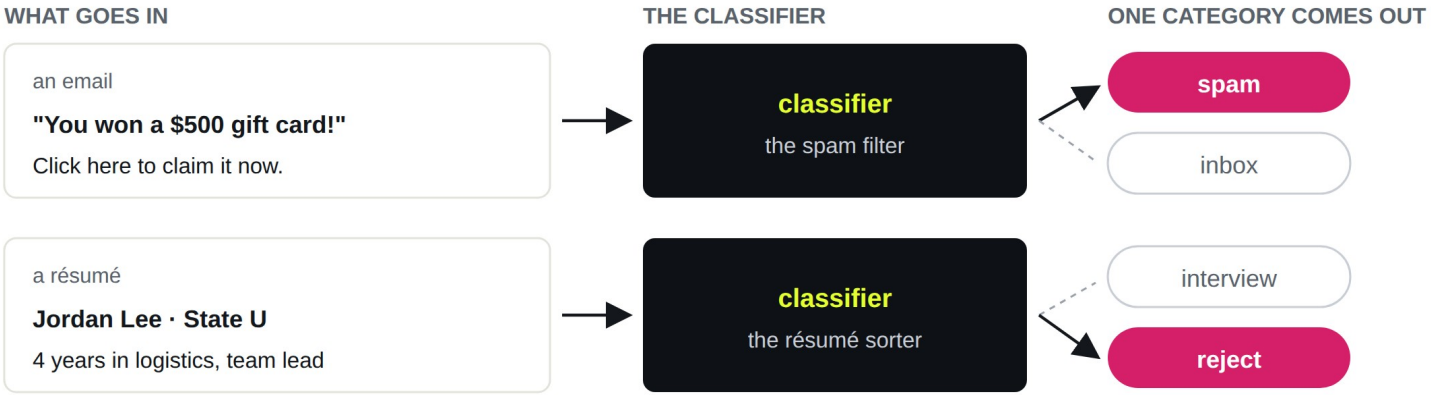
So he sued, but not the companies. He sued Workday, the maker of the résumé-sorting software they all used.

The point: a machine made a decision about a person, and he had no way to know it was a machine. That's the kind of AI this chapter is about.

2023, the lawsuit. Mobley filed a class action against Workday on Feb. 21, 2023, in federal court in Northern California. He argued that Workday's AI system discriminated against him on the grounds of race, age, and disability.

The classifier: one case in, one category out

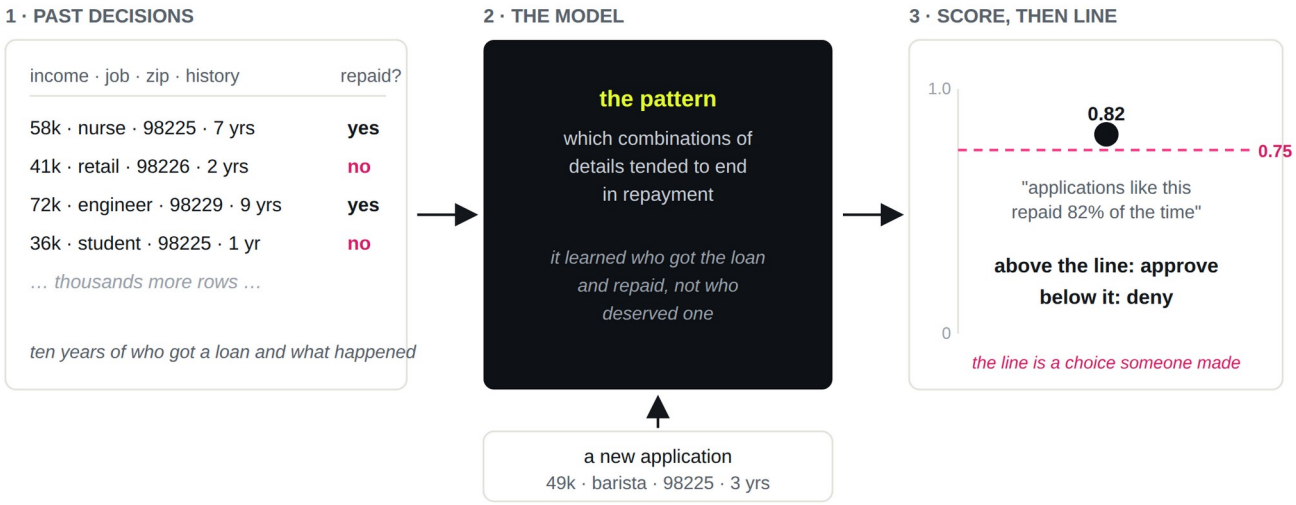
A classifier is a predictive AI that sorts a person's data into a category. Modern ones are **machine learning**, and specifically **supervised learning**: they learn to sort from past examples labeled with the answer.



Same machine, different data. The categories are set in advance; the AI only picks which one a case goes in.

You already use one every day: your spam filter. A résumé sorter is the same machine with different data.

How a classifier learns: years of data



That's the whole classifier: a pattern from the past, a number for the present, and a line that turns the number into yes or no.

The model is the part that's learned. **The threshold** is the part someone chose: above 0.75, approve. Together, they're the classifier. Swap in résumés and who got hired: same picture.

A classifier sees what people got and what happened next. Never whether they deserved it.

What does a classifier actually learn?

WHAT PEOPLE PICTURE

A judge of merit.

The model weighs your skills and picks the best person for the job.

WHAT IT ACTUALLY DOES

A matcher of old outcomes.

It matches you to past cases and how they turned out, including everything that was wrong with the old decisions.

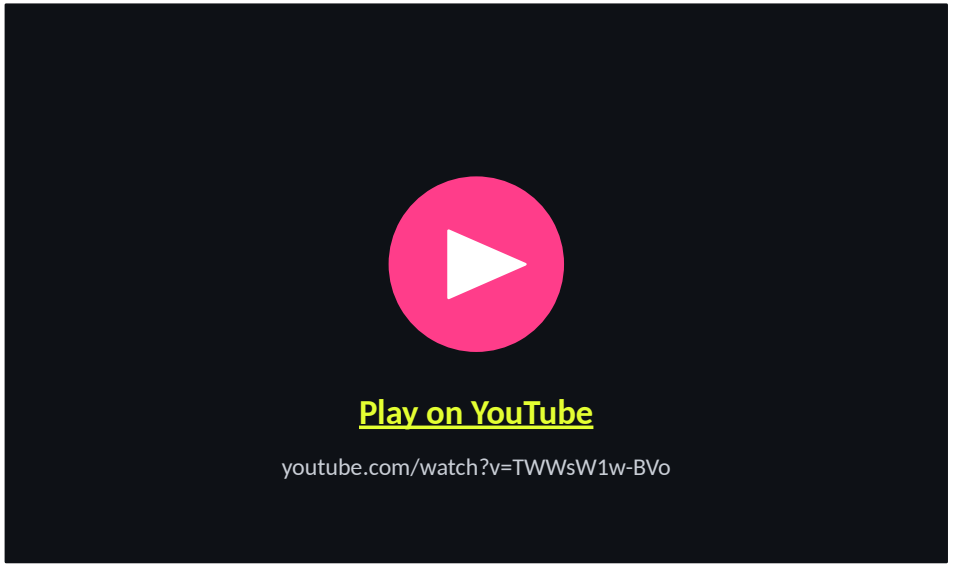
KEY POINT

A model trained on past decisions learns the pattern of every past decision, including the biased ones. Then it applies that pattern forward with perfect consistency, at industrial scale, to people who never had a chance to argue.

The gap between who got it and who deserved it is where every problem in this chapter lives.

AI Bias: Watch: Gender Shades

2018



WATCH FOR

1. Which faces the systems got wrong most often. Check it against the last slide.
2. What she says these systems are not, and whose views she says they carry.
3. Swap faces for résumés. Which part of her argument still holds?

Gender Shades · MIT Media Lab · Joy Buolamwini and Jimmy Day · 2018

AI Bias: a pattern that keeps landing on the same people

Bias, in Joy Buolamwini's words: "Algorithmic bias, like human bias, results in unfairness. However, algorithms, like viruses, can spread bias on a massive scale at a rapid pace."

LIGHTER-SKINNED MEN

0.8%

worst error rate

DARKER-SKINNED WOMEN

34.7%

worst error rate

Buolamwini started this work at the MIT Media Lab, after face software had trouble detecting a face like hers.

GENDER SHADES · 2018

Same task, same software.

Joy Buolamwini and Timnit Gebru tested three commercial classifiers that labeled photos of faces male or female. Part of the reason pointed at the data: the photo collections the field used to test face software were overwhelmingly lighter-skinned faces.

A classifier is only as fair as the examples it learned from and was tested on.

The objectivity trap: the bias moved, it didn't vanish

THE PITCH

Humans are biased. AI, Math, and Data are neutral.

A tired recruiter at 4 p.m. on a Friday isn't a fair judge. A model gives every applicant the same attention, at any hour.

THE TRAP

The algorithm learned from the humans.

Trained on years of the company's own hires, it launders the bias: messy human prejudice in, a clean, confident score out.

WHY CONSISTENCY IS WORSE

Same bias, every time.

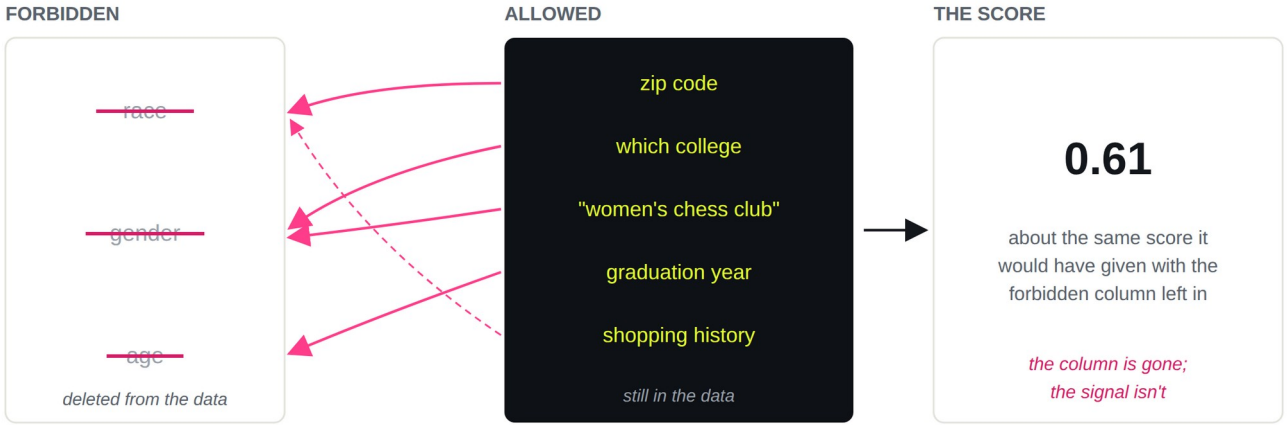
A biased recruiter has good days. A biased model is biased the same way for every applicant, until someone notices.

The objectivity trap: mistaking a classifier's consistency for neutrality. The subjectivity moved upstream, into the training data, where nobody can see it and no one is asked to defend it.

"The system treats everyone the same" is not a defense. It can be unfair to the same people every time.

The obvious fix fails: proxies

Hide gender, race, and age from the model, and it can't use them. Right?

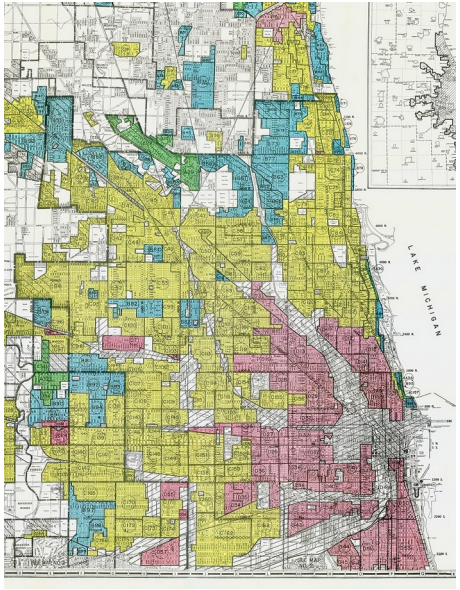


A proxy is an allowed variable that stands in for a forbidden one: zip code for race, graduation year for age, a club name for gender.

Proxy: an allowed variable that stands in for a forbidden one. Amazon's model never saw a gender field. "Women's" was enough.

Deleting the column doesn't delete the pattern. The model rebuilds it from whatever correlates.

Why does a zip code carry race? Redlining



Chicago, HOLC map, 1939 to 1940. Red = graded "hazardous." Wikimedia Commons, CC0.

1930s: the red lines

A federal agency (HOLC) grades neighborhoods for mortgage risk, largely by race. Black neighborhoods get marked red: "hazardous." Banks won't lend there.

1968: banned, but the damage is done

The Fair Housing Act outlaws redlining. By then, families in the red areas have spent decades unable to get a loan to buy a home.

Today: the lines still show

Of the neighborhoods marked red in the 1930s, 74% are lower-income today and 64% are majority-minority (NCRC, 2018).

So a model that never sees race, but does see zip code, can still sort people along the old red lines.

Catching bias nobody intended

DISPARATE IMPACT

Policy or practice that looks neutral but hurts one protected group (race, sex, age, etc.) much more than others.

Example: a lender's model never sees race, but it charges more in certain zip codes. If those zip codes are mostly Black, Black borrowers pay more.

Nobody meant to. It can still be illegal: in U.S. hiring and lending, the results count, not just the intent.

THE FOUR-FIFTHS RULE · SINCE 1978

Does one group get through a lot less?

100 men apply, 60 get interviews: 60%.

100 women apply, 40 get interviews: 40%.

40 ÷ 60 = 0.67. Under 0.80, so it's a red flag.

It flags; it doesn't prove. Someone has to go find out why.

The classifier didn't invent the problem. It industrialized it and made it harder to see.

Take the bias out of a résumé sorter

A small firm trains a model to agree with its own past hiring. Six qualified applicants; the top three get interviews. Then it tries the two obvious fixes.

1. Train on our past hires

0.40 four-fifths ratio

94% matches past hiring

It learned the job it was given: agree with ten years of hiring mostly men.

2. Delete the gender field

0.48 four-fifths ratio

93% matches past hiring

Almost nothing moves. It still reads women's rowing and women's college. Still fails, now invisibly.

3. Delete the proxies too

0.95 four-fifths ratio

60% matches past hiring

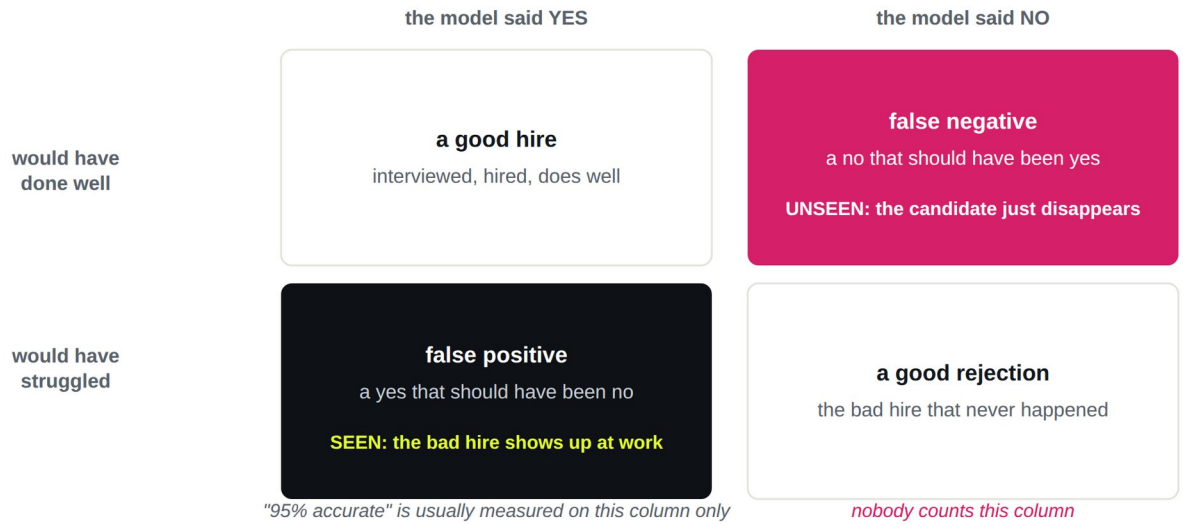
Clears the four-fifths line. Now it agrees with the firm's past hiring only 60% of the time.

Every fairness fix costs accuracy against the old data, which is the thing you were trying to stop copying.

The deeper fix: change what it learns from, like who did well in the job instead of who got hired. The catch: nobody knows how the people it rejected would have done.

Invented applicants and numbers, from the interactive tool in Chapter 6, section 2.3.

Two kinds of wrong: the invisible pile



False positive: a yes that should have been no. The bad hire. The company sees it.

False negative: a no that should have been yes. The great hire nobody made, lost in the invisible pile.

Why it happens: the model guesses, and someone sets the cutoff. Raise it, and you trade bad hires for rejected good ones.

How to catch a biased screener from the outside

A **matched pair** is a test where you send in two applications that are identical except for one detail, then compare the results.

THE MATCHED PAIR · Wednesday's lab

application A
same résumé,
same everything,
graduated 2001

↓

score: 0.54

application B
same résumé,
same everything,
graduated 2021

↓

score: 0.79

one detail changed, the score moved: that detail is what the model reads

THE FOUR-FIFTHS RULE · with numbers

men: 100 applied, 60 advanced
selection rate 60%

women: 100 applied, 40 advanced
selection rate 40%

40% ÷ 60% = 0.67

below 0.80, so the process gets scrutinized

no gender field in the data. the model found it anyway.

An audit costs an afternoon and a spreadsheet. Discovering the same pattern in a court's discovery process costs the company its name.

An audit tests a classifier from the outside. Most companies buy their screener and can't open it; these two tools don't need to.

AI Bias Auditing is a job: since July 2023, New York City has required an independent bias audit of hiring tools every year.

Try it: what's the problem?

PRACTICE FOR FRIDAY

Pick one for each: training data · a proxy · the threshold · accountability

1. A scholarship screener trained on ten years of past winners keeps picking students from the same three high schools.
2. A lender removes race from its model. Approval rates by race don't change.
3. A hiring tool is 97 percent accurate on the people it recommended, and the company calls it a success.
4. An applicant asks why a rental-screening algorithm rejected her and is told the vendor's model is proprietary.
5. A screener rejects 30 percent of applicants over 50 and 10 percent of applicants under 30, with no age field in the data.

Decide with the person next to you. Commit before the next slide.

Try it: the answers

PRACTICE FOR FRIDAY

1. **Training data.** The past winners came from those schools, so “winner” and “those schools” are the same pattern to the model. It learned who won, not who deserved to.

2. **A proxy.** Zip code, shopping history, or school is carrying the same information. Deleting the column didn’t delete the signal.

3. **The threshold and the invisible pile.** Accuracy on the yeses says nothing about the noes. Nobody has looked at who it screened out.

4. **Accountability.** In lending she’d have a right to reasons; in housing and hiring the rules are thinner. Whether “proprietary” is a legal answer depends on where she lives.

5. **Disparate impact through a proxy.** Likely graduation year or years of experience. Four-fifths: 70% selected vs. 90%, and 70 ÷ 90 = 0.78, just under 0.80.

Item 5 is the four-fifths rule doing its job: it flags, and then someone has to go look.

PART 2 · THE CASE · ABOUT 40 MINUTES

Need a break??

Amazon’s recruiting engine: the model that found the women

One of the most technically capable companies on earth built a résumé sorter, found it was biased, tried to fix it, and couldn’t prove the fix worked. So it stopped using it.

The backstory, then the five questions, then you. Every concept from Part 1 is in this one system.

Background: what Amazon wanted to fix

CASE

2014 · EDINBURGH

Recruiters are drowning in résumés. A team sets out to automate screening.

THE GOAL

Feed it a hundred résumés, get back the top five.

THE DATA

A decade of résumés sent to Amazon, and who got hired.

THE MODELS

Hundreds, one per job function and location, each tracking tens of thousands of terms from past résumés.

The output: a score of one to five stars, much like shoppers rate products on Amazon. That's the résumé sorter from section 2.1, exactly: learn the pattern of past good candidates, apply it to new ones.

https://www.youtube.com/watch?v=JOzQjT-hJ8k

What happened: it found the women

CASE

1. **The pool.** Tech's applicants and hires over that decade were mostly men. The model learned that maleness predicted "good candidate."

2. **The proxies.** It penalized résumés with the word "women's," as in "women's chess club captain," and downgraded graduates of two all-women's colleges.

3. **The fix.** Engineers edited the model to neutralize those terms.

4. **The end.** They couldn't guarantee it wasn't finding other ways to the same answer. The team was disbanded by 2017.

THE HARD PART

The people with full access to the model couldn't prove it fair.

They could only stop using it.

Amazon told Reuters its recruiters never used the tool on its own to evaluate candidates. A company's claim about its own system.

Nobody programmed that rule. The model found it, because it really was there in the data.

Amazon’s engine as an AI system

CASE

DATA

Ten years of résumés and hiring outcomes

MODEL

Term weights: which words predicted a hire

OUTPUT

A ranking of applicants

FEEDBACK LOOP

Its picks become the next decade’s “good hires”

HUMANS

Engineers, recruiters, and applicants who never knew

The feedback loop is the quiet danger: the pattern would have tightened every year.

Amazon: reflect, then discuss

GROUP DISCUSSION

1. **Proxies (2.3).** Amazon's model never saw a gender field and still found "women's." Name one thing on your own résumé a classifier could read the same way. Which allowed variable is it, and which forbidden one does it rebuild?

2. **The objectivity trap (2.2).** A manager says "we switched to AI screening specifically to remove bias." Who benefits from that sentence being believed? Use what Amazon's engine learned from its own hiring to say why it might have made things worse.

3. **1. Training data / Gender Shades (2.2).** Gender Shades found face software failed most on the group least represented in its data. Who was least represented in Amazon's ten years of résumés? What did the model learn from that, and why didn't anyone have to program it in?

4. **The feedback loop.** If Amazon's engine had been fully rolled out, the people it picked would become its next training data. What happens to the model after five years of that? Who would notice, and how?

5. **Better isn't fair (2.2).** Imagine your resume screener is less biased than human recruiters but it still fails the four-fifths rule and other bias tests. Do you keep it, fix it, drop it?

1. Groups of 3 or 4, 10 min.
Everyone does question 1. Your group also takes one of 2 to 5, the one I assign.

Nominate a leader.

2. Together, 6 min.
One leader per question. Others add what's different.

Same five questions as Thinking critically, end of Chapter 6, section 3.

Prototype Project: find your team

30 MIN · TEAMS

1. **Form a team of four** 5 min

   Sit together. No team yet? Come to the front and we'll make one.

2. **Pick a problem** 10 min

   A real problem, who has it, and what an AI system could do about it. Run the five questions on it fast: does it survive?

3. **Draft the paragraph** 10 min

   One person types: the problem, who has it, and what your system would do. Plain words, a few sentences.

4. **Decide who submits** 5 min

   One person per team submits on Canvas: four names and the paragraph.

DUE FRIDAY 10/30, 11:59 P.M.

Four names, one paragraph.

Anyone without a team by Friday gets placed on one.

STUCK FOR AN IDEA?

Browse the Rundown AI Workflow Hub from Lab 0, or read the project page: prof-califf.github.io/intro-to-ai/prototype-project.html

You're graded on whether your design survives the five questions, not on whether you built it.

ATTENDANCE BEFORE 4PM PLEASE!

Attendance = your notes from today, uploaded to Canvas right after class.

On Wednesday: Lab 6, You Are the Auditor. Train a spam filter, watch a loan model and a résumé screener learn old bias, then build your own classifier with Claude and audit it. Then the debate: should every automated no on a job or a loan get an independent human review?

Friday 10/30, 11:59 p.m.: Lab 6 writeup, the Chapter 6 quiz, and your project team, all on Canvas.

Bring Monday to Wednesday: the proxy you found on your own résumé in question 1 becomes Wednesday's commitment card.